MATH 420. Homework 3. Sadullaev L.

Problem 1.

Determine if the set S is L.I:

(a)

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ -4 & 4 \end{bmatrix} \right\}$$

$$a \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} + c \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 2 & 1 \\ -4 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad (1)$$

$$\begin{cases} a + b\cdot 0 + c\cdot(-1) + d\cdot 2 = 0 \\ a\cdot 0 + b(-1) + c\cdot 2 + d = 0 \\ a\cdot(-2) + b(1) + c\cdot 1 + d(-4) = 0 \\ a(-1) + b\cdot(1) + c\cdot 0 + d\cdot(4) = 0 \end{cases} \Rightarrow \begin{cases} a - c + 2d = 0 \\ -b + 2c + d = 0 \\ -2a + b + c - 4d = 0 \\ a + b + 4d = 0 \end{cases} \begin{matrix} \} - \\ \\ \} + \\ \\ \end{matrix}$$

$$\begin{cases} a + 2b - 5c = 0 \\ -a + 2b + c = 0 \\ 4b - 4c = 0 \end{cases} \quad \begin{cases} a = c - 2d = c - 2(b - 2c) \\ d = b - 2c \\ b = c \end{cases} \quad \forall c \in \mathbb{R}$$

(1) has nontrivial solution, so

S is L.D

(b) $S = \left\{ \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix} \right\}$ (2)

$$a \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} + c \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (2)$$

$$\begin{cases} a - c + 2d = 0 \\ -2b + 4c + 2d = 0 \\ -2a + b + c + 2d = 0 \\ a + b - 2d = 0 \end{cases} \quad \begin{cases} a + 2b - 5c = 0 \\ -a + 2b + c = 0 \\ 4b - 4c = 0 \end{cases} \quad \boxed{b = c}$$

$a = 5c - 2b = 3$ $\quad \forall c = 1.$

$d = b - 2c = -1$

(2) has nontrivial solution $\Rightarrow$

S is not L.I.

(c) $S = \{ x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1 \}$

$$a(x^3 + 2x^2) + b(-x^2 + 3x + 1) + c(x^3 - x^2 + 2x - 1) = 0 \quad (3)$$

$$\begin{cases} a + c = 0 \quad a = -c \\ 2a - b - c = 0 \\ 3b + 2c = 0 \Rightarrow b = \frac{-2c}{3} \\ b - c = 0 \Rightarrow b = c \end{cases} \quad \Big\} -\frac{2}{3}c = c \Rightarrow c = 0 \quad \cdot\ a = 0$$

(3) has trivial solution $\Rightarrow$ S is L.I.

(d) $S = \{x^3 - x,\ 2x^2 + 4,\ -2x^3 + 3x^2 + 2x + 6\}$

$$a(x^3 - x) + b(2x^2 + 4) + c(-2x^3 + 3x^2 + 2x + 6) = 0 \quad (4)$$

$$\begin{cases} a - 2c = 0 \\ 2b + 3c = 0 \\ a + 2c = 0 \\ 4b + 4c = 0 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = 0 \\ c = 0 \end{cases}$$

(4) has <u>only</u> trivial solution $\Rightarrow$

S is L.I. $\square$

Problem 2.

Let S be a set of <u>nonzero</u> polynomials in P, s.t. no two have the same degree. Prove that S is L.I.

Solution:

Let S be a set of nonzero polynomials in P, s.t

$$\text{degree}(p) \neq \text{degree}(q)$$

for $p, q \in S$

For contradiction, assume S is L.D.
Then based on the definition of L.D. sets
$\exists$ <u>finite number</u> of vectors $p_1, p_2 \dots p_n \in S$
and scalars $a_1, a_2 \dots a_n \in F$, <u>not all zero</u>, s.t.

(1) $a_1 p_1 + a_2 p_2 + \dots + a_i p_i(x) + \dots a_j p_j(x) + \dots + a_n p_n = 0$

Note that degree $(p_i(x)) \neq$ degree $(p_j(x))$ (2)
for $i \neq j$, $i, j = 1, 2 \dots n$. $\Rightarrow$

WLOG, for convinience let's say
degree $(p_i(x)) = i$. $\Rightarrow$

Then based on (2), there is no $a_j x^j$ term which I can group it with $a_i x^i$ to make $(a_i + a_j) x^i$ since $x^i \neq x^j$ for $i \neq j$.
That means my $a_i x^i$ term appears in (1) <u>once only</u> which means $a_i x^i$ term never disappears by cancelling with other

like terms, since we do not have any like terms to x^i. So to erase $a_i x^i$ term in (1), I must have $a_i = 0$, and this business holds for all a_i terms for $i = 1, 2 \ldots n$. $\Rightarrow$

$$a_1 = a_2 = \ldots = a_n = 0 \qquad (3)$$

Which is a contradiction for our assumption that S is L.D, so not all a_i scalars are zero. $\Rightarrow$ S is L.I. $\square$

Problem 3.

Prove that if S is a L.I. subset of $M_{n\times n}(\mathbb{R})$, then $S^T = \{A^T \mid A \in S\}$ is also L.I.

Solution

Let S be a L.I. subset of $M_{n\times n}(\mathbb{R})$. (1')

Since we're on $M_{n\times n}(\mathbb{R})$, for sure

S must be finite set, and let S be

$$S = \{A_1, A_2, \dots A_m \mid A_i \in M_{n\times n}(\mathbb{R})\} \quad (2')$$

Then $\exists$ scalars $\alpha_1, \dots \alpha_n$ $\ni$

$$\alpha_1 A_1 + \alpha_2 A_2 + \dots \alpha_m A_m = 0 \quad (3').$$

Since S is L.I., (3) has only $(\alpha_1, \alpha_2, \dots \alpha_n) = (0, 0, \dots 0)$ solution. (4')

By the fact $(A+B)^T = A^T + B^T$ (5'), let's transpose (3):

$$(\alpha_1 A_1 + \alpha_2 A_2 + \dots + \alpha_m A_m)^T = [0]^T \quad (6') \Rightarrow$$

$$\alpha_1 A_1^T + \alpha_2 A_2^T + \dots + \alpha_m A_m^T = 0 \quad (6')$$

I need to show that the <u>only solution</u> for (6') again is $(0, \dots 0)$.

Let's try to build a system of linear

equations for $(\alpha_1 .. \alpha_n)$ using (3')

$$\begin{cases} \alpha_1 (A_1)_{11} + \alpha_2 (A_2)_{11} + \dots + \alpha_m (A_m)_{11} = 0 & (1) \\ \alpha_1 (A_1)_{12} + \alpha_2 (A_2)_{12} + \dots + \alpha_m (A_m)_{12} = 0 & (2) \\ \dots\dots\dots\dots\dots\dots & \\ \alpha_1 (A_1)_{nn} + \alpha_2 (A_2)_{nn} + \dots + \alpha_m (A_m)_{nn} = 0 & (n^2) \end{cases}$$

We have n^2 equations from all entries of $(A_p)_{ij}$ $(p = 1, 2 .. m,\ i, j = 1, 2 .. n)$ for only $(\alpha_1 .. \alpha_m)$ unknowns. (7')

If we try to build such kind of linear system for $(\alpha_1 .. \alpha_m)$ unknowns using (6'), since we have to take again all entries of $(A_p^T)_{ij}$ we will come up with the same (1), (2) .. (n^2) equations. The reason for this is

$(A^T_P)_{ij} = (A_P)_{ji}$ which means the same coefficients makes up the ith equation for $(A^T_P)_{ij}$ while we have the same equation on the jth line of $(A_P)_{ji}$. So

$$\alpha_1 = \alpha_2 = \dots \alpha_n = 0 \qquad (8')$$

is the only solution for $(6')$.

$\Rightarrow S^+$ is L.I.

$\square$

Problem 4.

Let S_1 & S_2 be disjoint L.I. subsets of a vector space V. Prove that $S_1 \cup S_2$ is L.D. if and only if

$$\text{span}(S_1) \cap \text{span}(S_2) \neq \{0\}$$

Solution:

($\Rightarrow$)

Let S_1 & S_2 be (1) disjoint (2) linearly independent (L.I.) subsets of a vector space V. $\Rightarrow$

$$S_1 \cap S_2 = \emptyset \qquad (1)$$

Assume $S_1 \cup S_2$ is L.D. (3)

Then WTS: $\text{span}(S_1) \cap \text{span}(S_2) \neq \{0\}$

Assume for contradiction $\text{span}(S_1) \cap \text{span}(S_2) = \{0\}$ (4)

If $S_1 \cup S_2$ is L.D., then $\exists$ finite number of vectors $x_1 \ldots x_m \in (S_1 \cup S_2)$ and

scalars $a_1, a_2 \ldots a_m \in F$, not all zero $\ni$

$$a_1 x_1 + a_2 x_2 + \ldots + a_m x_m = 0$$

Recall that $S_1 \cap S_2 = \emptyset$.

Then $\forall\, v_1 \ldots v_n \in S_1$ & scalars $a_1 \ldots a_n \in F$ (including some nonzero and zero, but not all zero)

and

$\forall\, u_1 \ldots u_m \in S_2$ & scalars $b_1 \ldots b_m$ (not all zero)

$$a_1 v_1 + a_2 v_2 + \ldots + a_n v_n \neq b_1 u_1 + \ldots + b_m u_m \quad (5)$$

because of (4) on page 9.

$$\Rightarrow a_1 v_1 + a_2 v_2 + \ldots + a_n v_n - (b_1 u_1 + \ldots + b_m u_m) \neq 0 \quad (6)$$

for some $a_i \neq 0$ & some $b_j \neq 0$ $\Rightarrow$

$\Rightarrow \{v_1 \ldots v_n, u_1 \ldots u_m\}$ is L.I.

But it is a contradiction. $\{v_1 \ldots v_n, u_1 \ldots u_m\}$ should be L.D instead because of (3). So, $\mathrm{span}(S_1) \cap \mathrm{span}(S_2) \neq \{0\}$